VIT

Vellore Institute of Technology

CHENNAI

Reg. Number:

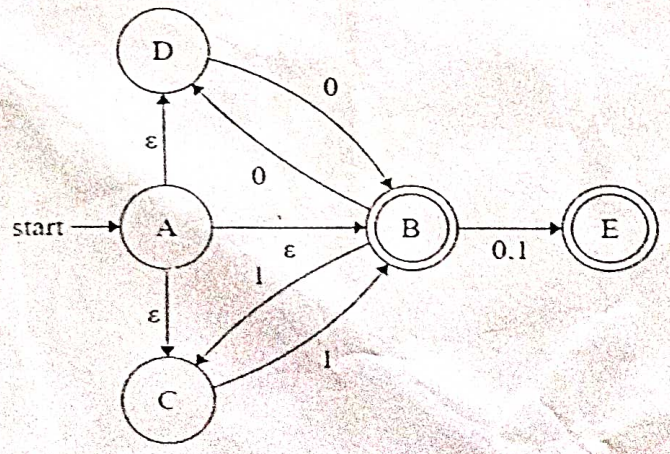
Continuous Assessment Test (CAT) – 1 AUGUST 2024

Programme	:	BTech Computer Science and Engineering AI-ML	Semester	:	FALL 24-25
Course Code & Course Title	:	BCSE304L THEORY OF COMPUTATION	Class Number	:	CH2024250100534, CH2024250101447, CH2024250100912
Faculty	:	Dr. S Kiruthika, Dr. Anand M, Dr. Nathezhtha T	Slot	:	A2+TA2
Duration	:	90 MINUTES	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information.
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Description	Marks
1.	a) Construct the deterministic finite automaton for a language over the $\Sigma = \{a,b\}$, which accepts all strings that begin and end with different symbol. (5 Marks) b) Construct the deterministic finite automaton for a language over the $\Sigma=\{0,1\}$, for all strings in which the number of zeros is divisible by 4. (5 Marks)	10
2.	Convert the given ε -NFA into NFA without ε moves with necessary steps.  	10
3.	Construct the finite automaton for the following languages a) $L_1 = \{w \mid w \in \Sigma = \{a,b\}^*$, where w contains the strings starting with	10

aabb}. (2.5 Marks)

b) L_2 = {w | w∈ ∑= {a,b,c}*, where w contains strings where third symbol from right is *a*, second symbol from right is *b* and first symbol from right is *c*}. (2.5 Marks)

c) L_3 = {w | w ∈∑= {a,b}*, where w contains the strings which ends with two consecutive *a*'s or two consecutive *b*'s}. (2.5 Marks)

d) L_4 = {w | w ∈∑ ={a,b}*, where w does not contains the strings which ends with *ab*}. (2.5 Marks)

4. You are designing a search engine feature that filters the queries based on certain patterns. Users often input the sequences of characters, and your system needs to determine whether a query is valid based on specific rules. The valid queries are either: Any combination of '*a*' and '*b*' (including empty strings), or multiple occurrences of the pattern "*ba*". Additionally, the valid query must end with one or more occurrences of a string starting with '*a*', followed by zero or more '*b*'s, and ending with '*a*'. For this scenario construct a DFA through ε -NFA 10

5. How can a given DFA be optimized to have the fewest possible states while preserving its language recognition capabilities? 10

M= ({q0, q1, q2, q3, q4, q5, q6, q7}, {0,1}, δ (refer table), q0 (initial state), q2 (final state))

δ (state, input)	0	1
→q0	q1	q5
q1	q6	q2
q2*	q0	q2
q3	q2	q6
q4	q7	q5
q5	q2	q6
q6	q6	q4
q7	q6	q2

**********All the best **********